

MODULE 03

SQL Data Types and Constraints

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

SQL Data Types and Constraints

Numeric Types

INT: whole numbers

DECIMAL(p,s): exact decimals e.g. salary DECIMAL(10,2)

FLOAT: approximate floating-point

BIGINT: very large integers

String Types

CHAR(n): fixed-length string

VARCHAR(n): variable-length (most common)

TEXT: large text blocks

NVARCHAR: unicode variable-length

Date and Time Types

DATE: YYYY-MM-DD

TIME: HH:MM:SS

DATETIME: date and time combined

TIMESTAMP: with timezone support

Constraints

NOT NULL: column cannot be empty

UNIQUE: all values must differ

PRIMARY KEY: NOT NULL + UNIQUE

FOREIGN KEY: links to another table

CHECK: validates a condition

DEFAULT: sets a fallback value

Full Example

```
CREATE TABLE employees (  
  id INT PRIMARY KEY,  
  name VARCHAR(100) NOT NULL,  
  email VARCHAR(150) UNIQUE,  
  salary DECIMAL(10,2) CHECK (salary > 0),  
  dept_id INT REFERENCES departments(id)  
);
```